TERRAFORM-01

Task

Objective: Install Terraform on the local desktop and configure it for infrastructure automation.

Steps Performed:

1. Downloaded and installed Terraform on the Windows desktop.
2. Configured Terraform in the system PATH.
3. Opened the terminal and verified the installation using:

   ```
   terraform -version
   ```

4. Confirmed Terraform was available from the command line.

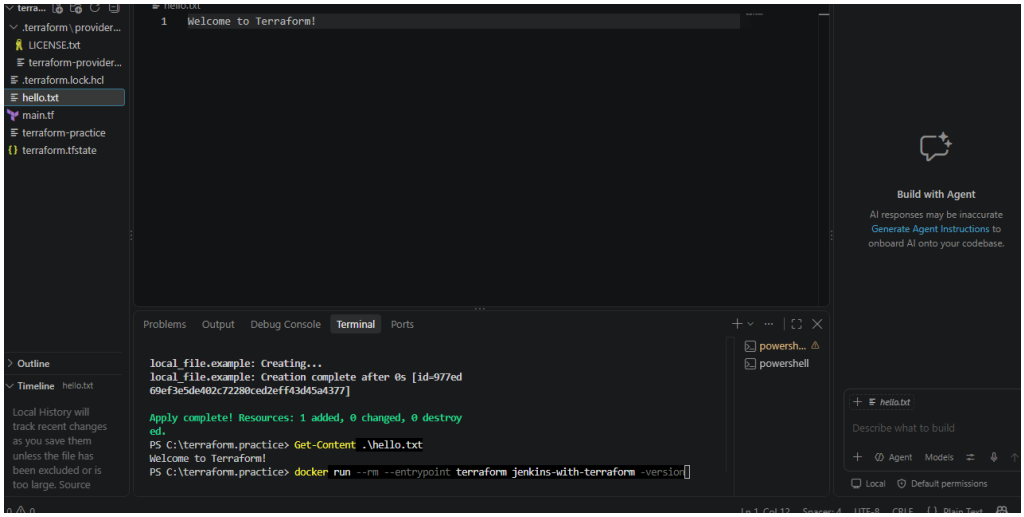
Result: Terraform was installed and verified successfully on the desktop.

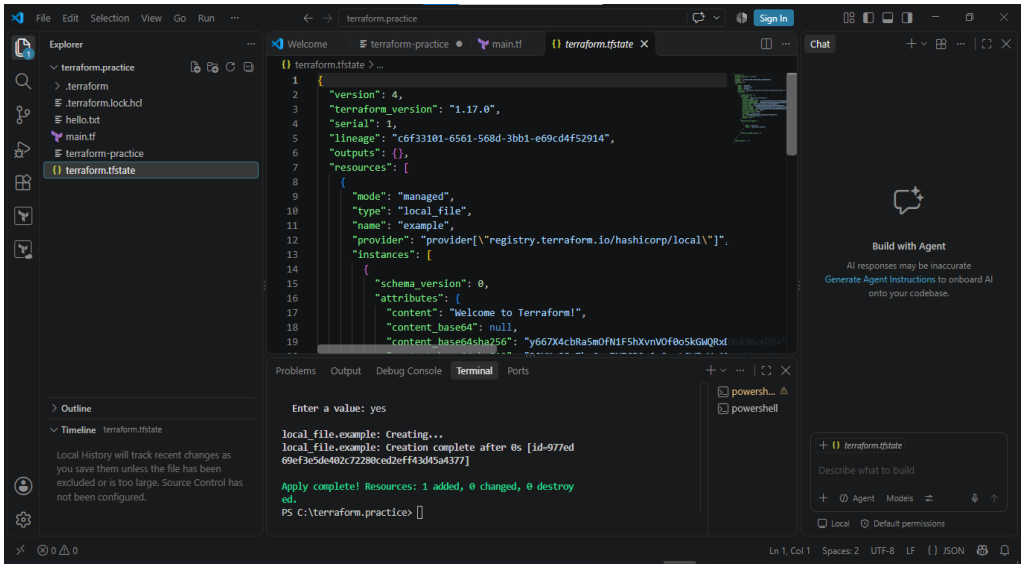


Task:-

Objective: Understand the purpose of Terraform commands and providers used to manage infrastructure.

Concept	Purpose
`terraform init`	Initializes the working directory and downloads required provider plugins.
`terraform plan`	Previews the infrastructure changes Terraform will make.
`terraform apply`	Executes the plan and creates or updates the defined resources.
`Terraform Provider`	A plugin that allows Terraform to interact with a platform or service, such as AWS, Azure, or local files.

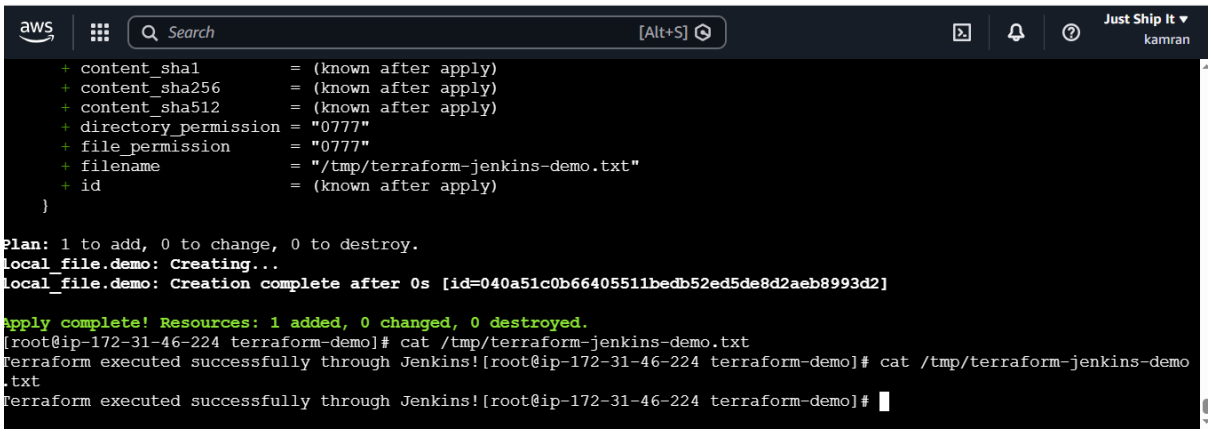


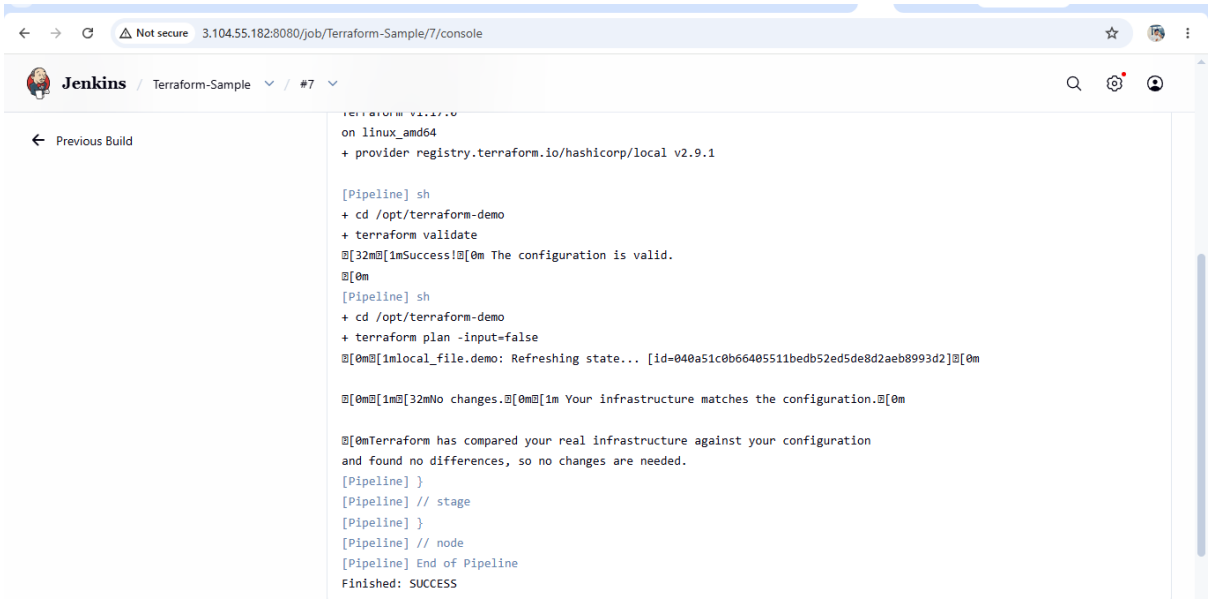


Task: Terraform Integration with Jenkins

Objective: Automate Terraform execution through a Jenkins pipeline.

1. Installed Terraform on the Jenkins EC2 instance.
2. Created a sample Terraform configuration using the `local_file` resource.
3. Configured a Jenkins Pipeline job named `Terraform-Sample`.
4. Executed Terraform validation and plan stages through Jenkins.
5. Verified the pipeline completed successfully.




Not secure
3.104.55.182:8080/job/Terraform-Sample/7/console
Jenkins
Terraform-Sample
#7
Previous Build
on linux_amd64
+ provider registry.terraform.io/hashicorp/local v2.9.1
[Pipeline] sh
+ cd /opt/terraform-demo
+ terraform validate
[32m[1mSuccess![0m The configuration is valid.
[0m
[Pipeline] sh
+ cd /opt/terraform-demo
+ terraform plan -input=false
[0m[1mlocal_file.demo: Refreshing state... [id=040a51c0b66405511bedb52ed5de8d2aeb8993d2][0m
[0m[1m[32mNo changes.[0m[1m Your infrastructure matches the configuration.[0m
[0mTerraform has compared your real infrastructure against your configuration
and found no differences, so no changes are needed.
[Pipeline] }
[Pipeline] // stage
[Pipeline] }
[Pipeline] // node
[Pipeline] End of Pipeline
Finished: SUCCESS